



Electronics I

Lecture 1

Sinusoidal wave form

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Course Details:

- **Textbook:**

1- Electronic Principles, A. Malvino and D. Bates, 8th Ed. 2015.

2- Electronic Devices and Circuit Theory, R. L. Boylestad, L. Nashelsky, 11th Ed. 2013

3 - Solid State Electronic Devices, K. Bhattacharya and R. Sharm, 2nd Ed. 2013.

4- University Physics with Modern Physics, H. D. Young and R. A. Freedman, 14th Ed. 2016.

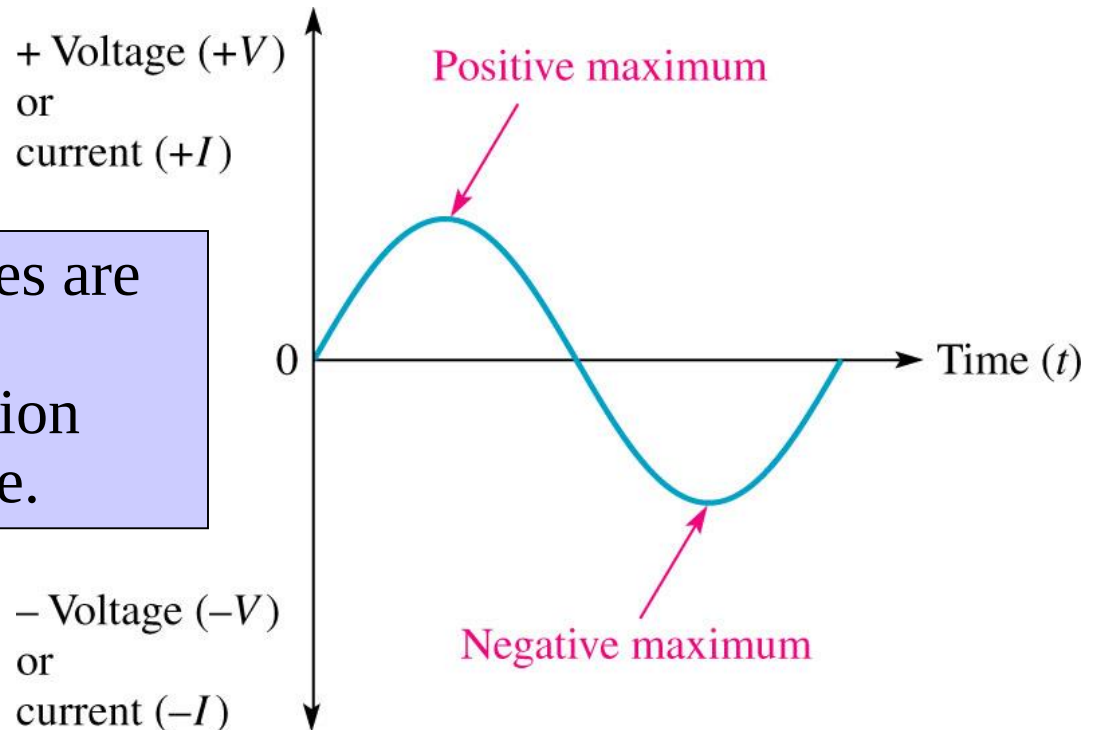
- **Course Assessment:**

- Labs: **10%**
- Midterms and Exams: **30%**
- Final: **60%**

Sine waves

The sinusoidal waveform (sine wave) is the fundamental alternating current (ac) and alternating voltage waveform.

Electrical sine waves are named from the mathematical function with the same shape.



A wave is a disturbance. Unlike water waves, electrical waves cannot be seen directly but they have similar characteristics. All periodic waves can be constructed from **sine waves**, which is why sine waves are fundamental.

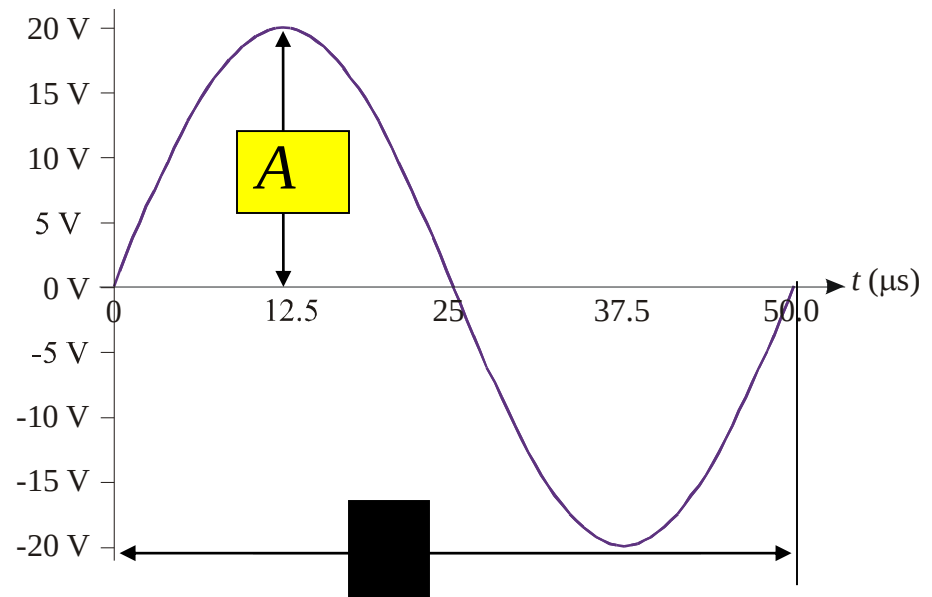


Sine waves

Sine waves are characterized by the amplitude and period. The **amplitude** is the maximum value of a voltage or current; the **period** is the time interval for one complete cycle.

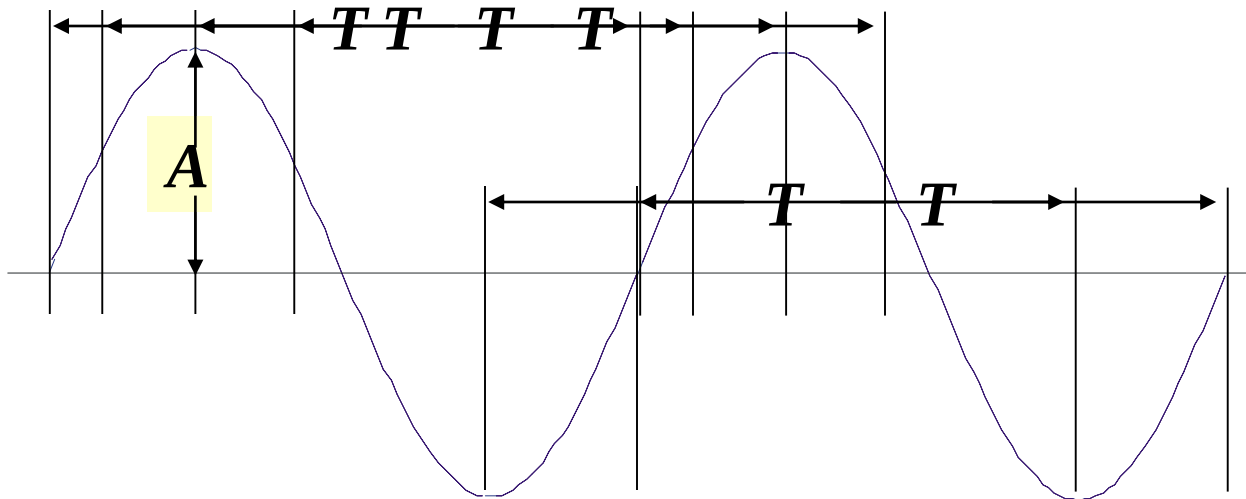
The amplitude (A) of this sine wave is **20 V**

The period is **50.0 μs**



Sine waves

The period of a sine wave can be measured between any two corresponding points on the waveform.



By contrast, the amplitude of a sine wave is only measured from the center to the maximum point.

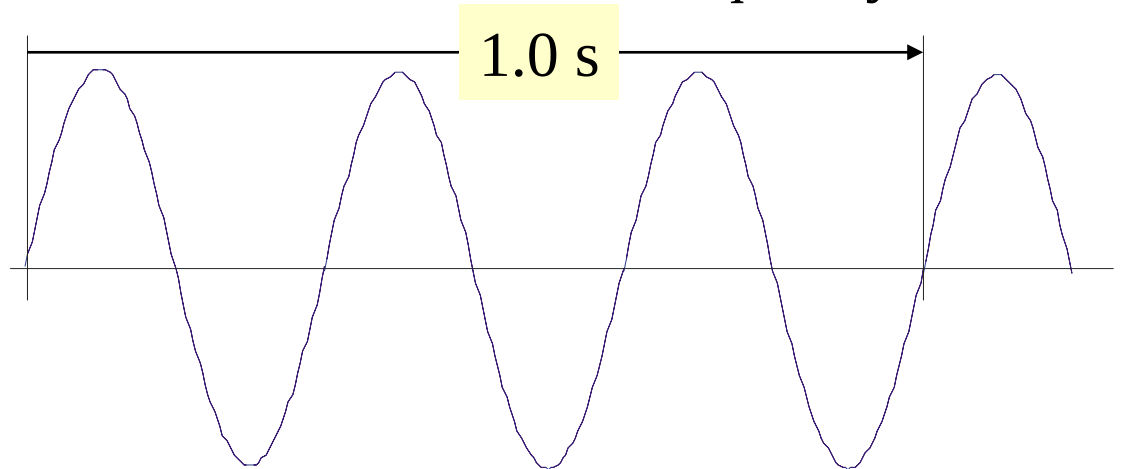
Summary

Frequency

Frequency (f) is the number of cycles that a sine wave completes in one second.

Frequency is measured in **hertz** (Hz).

If 3 cycles of a wave occur in one second, the frequency is **3.0 Hz**



Period and frequency

The period and frequency are reciprocals of each other.

$$f = \frac{1}{T}$$

and

$$T = \frac{1}{f}$$

Thus, if you know one, you can easily find the other.

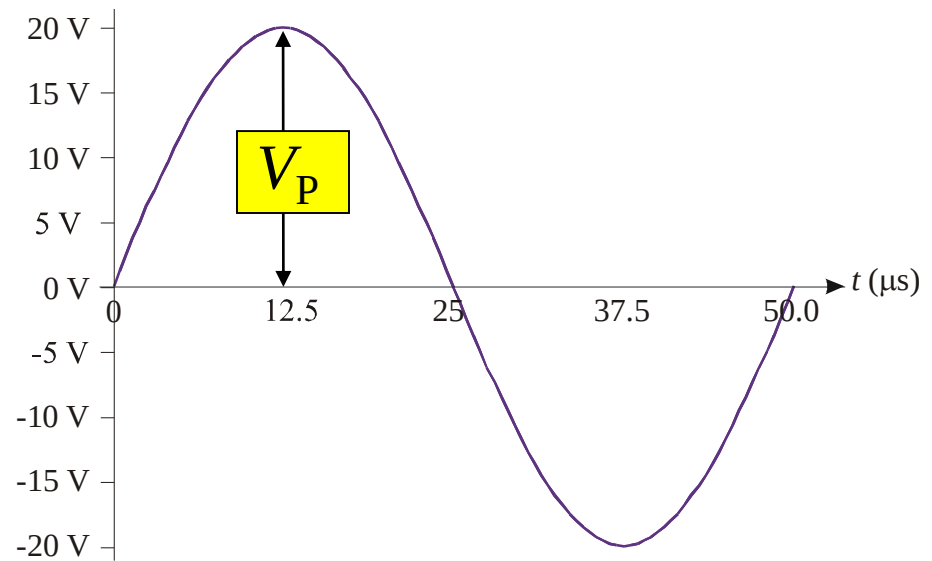
(The $1/x$ key on your calculator is handy for converting between f and T .)

If the period is $50\ \mu\text{s}$, the frequency is $0.02\ \text{MHz} = 20\ \text{kHz}$.

Sine wave voltage and current values

There are several ways to specify the voltage of a sinusoidal voltage waveform. The amplitude of a sine wave is also called the peak value, abbreviated as V_P for a voltage waveform.

The peak voltage of this waveform is **20 V**.

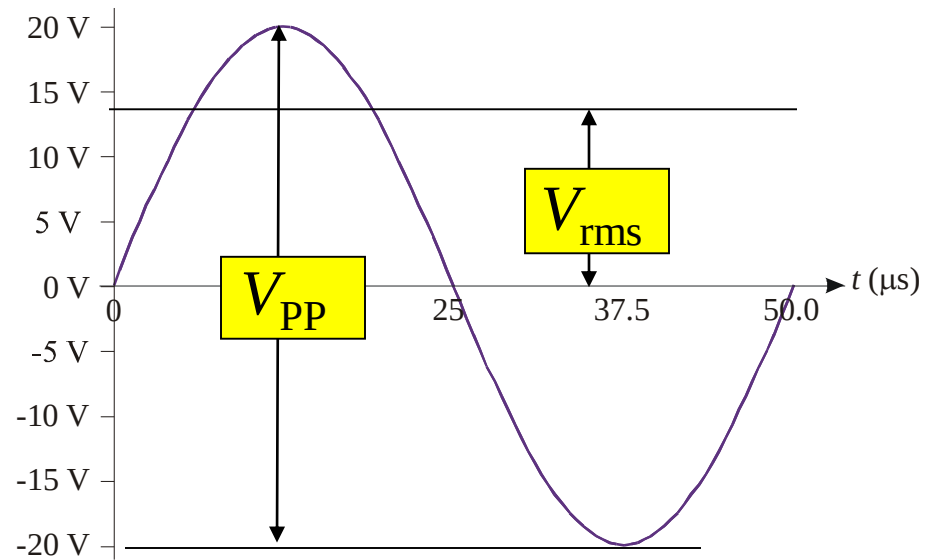


Sine wave voltage and current values

The voltage of a sine wave can also be specified as either the peak-to-peak or the rms value. The peak-to-peak is twice the peak value. The rms value is 0.707 times the peak value.

The peak-to-peak voltage is **40 V**.

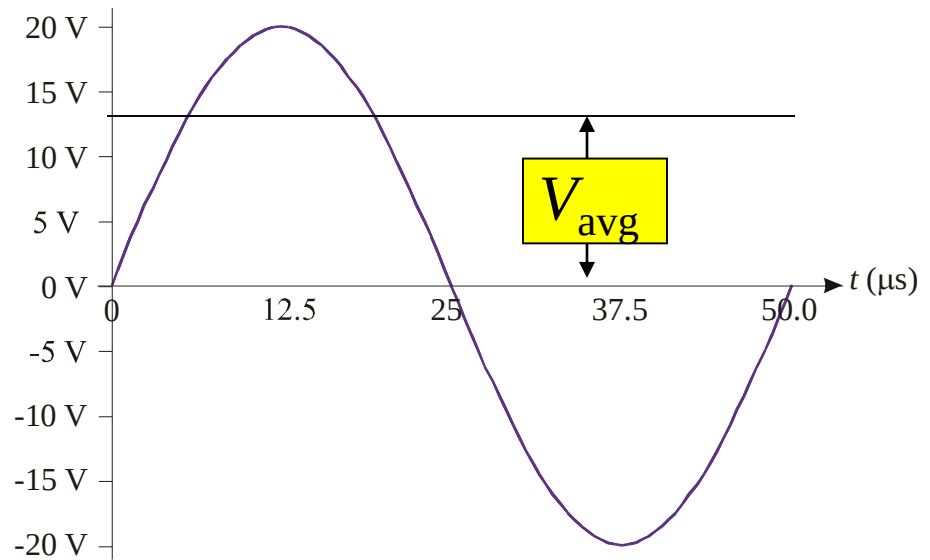
The rms voltage is **14.1 V**.



Sine wave voltage and current values

For some purposes, the average value (actually the half-wave average) is used to specify the voltage or current. By definition, the average value is as 0.637 times the peak value.

The average value for the sinusoidal voltage is **12.7 V**.

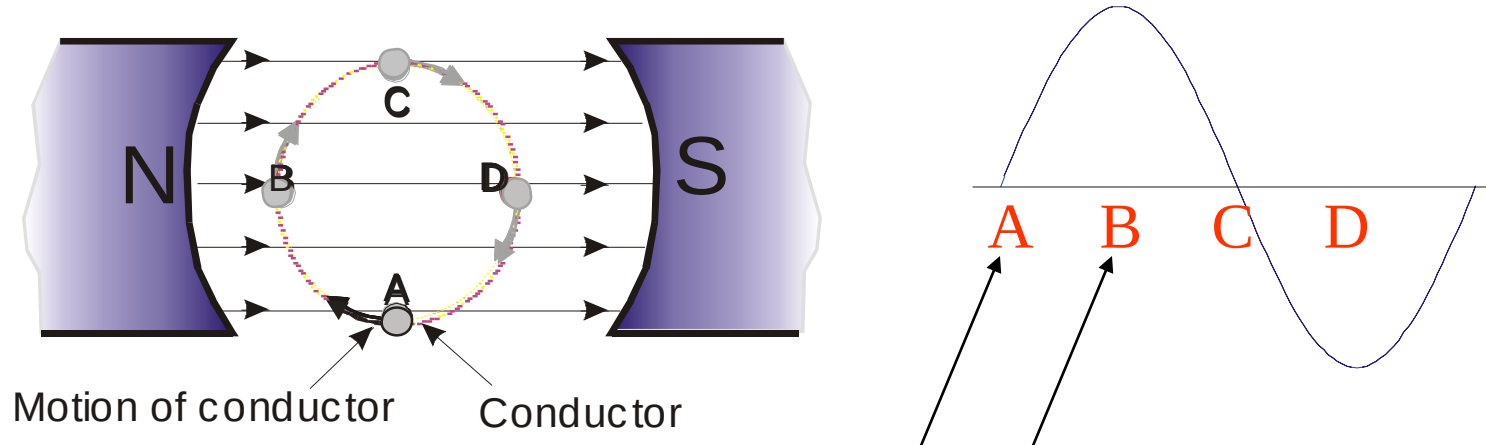


Summary

Generation of a sine wave

Sinusoidal voltages are produced by ac generators and electronic oscillators.

When a conductor rotates in a constant magnetic field, a sinusoidal wave is generated.

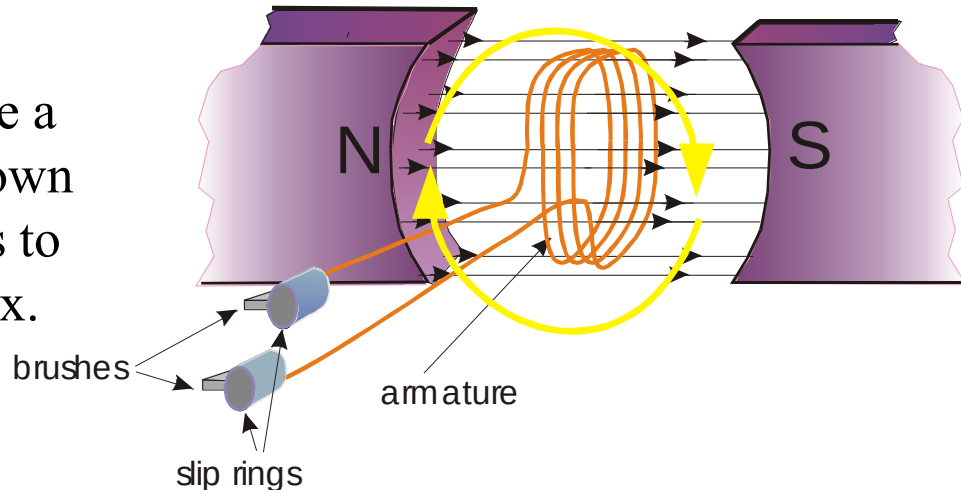


When the conductor is moving parallel with the lines of flux, no voltage is induced.

AC generator (alternator)

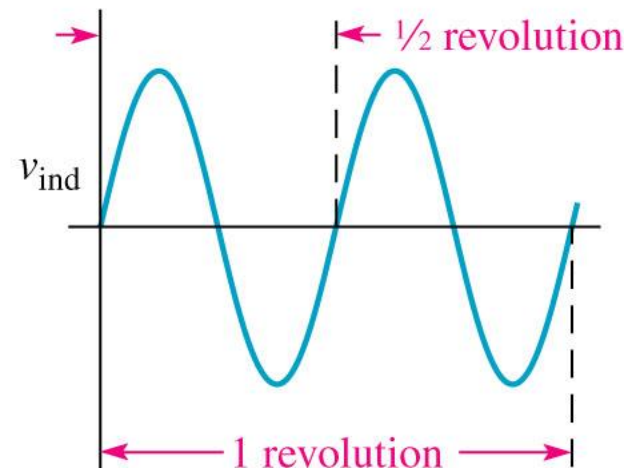
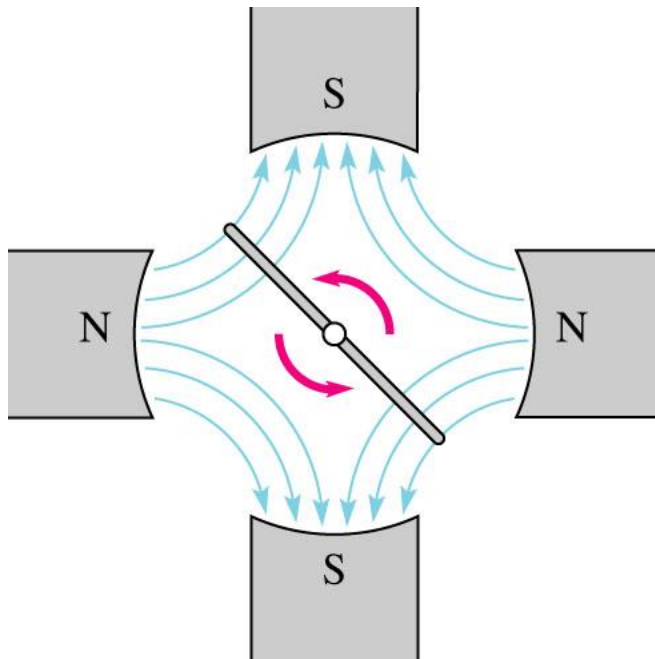
Generators convert rotational energy to electrical energy. A stationary field alternator with a rotating armature is shown. The armature has an induced voltage, which is connected through slip rings and brushes to a load. The armature loops are wound on a magnetic core (not shown for simplicity).

Small alternators may use a permanent magnet as shown here; other use field coils to produce the magnetic flux.



AC generator (alternator)

By increasing the number of poles, the number of cycles per revolution is increased. A four-pole generator will produce two complete cycles in each revolution.



Selected Key Terms

Sine wave A type of waveform that follows a cyclic sinusoidal pattern defined by the formula $y = A \sin \theta$.

Alternating current Current that reverses direction in response to a change in source voltage polarity.

Period (T) The time interval for one complete cycle of a periodic waveform.

Frequency (f) A measure of the rate of change of a periodic function; the number of cycles completed in 1 s.

Hertz The unit of frequency. One hertz equals one cycle per second.

Selected Key Terms

- Instantaneous value*** The voltage or current value of a waveform at a given instant in time.
- Peak value*** The voltage or current value of a waveform at its maximum positive or negative points.
- Peak-to-peak value*** The voltage or current value of a waveform measured from its minimum to its maximum points.
- rms value*** The value of a sinusoidal voltage that indicates its heating effect, also known as effective value. It is equal to 0.707 times the peak value. *rms* stands for root mean square.

Selected Key Terms

Radian A unit of angular measurement. There are 2π radians in one complete 360° revolution.

Phasor A representation of a sine wave in terms of its magnitude (amplitude) and direction (phase angle).

Amplitude The maximum value of a voltage or current.

Pulse A type of waveform that consists of two equal and opposite steps in voltage or current separated by a time interval.

Harmonics The frequencies contained in a composite waveform, which are integer multiples of the pulse repetition frequency.